Quantifying Cluster Compactness vs. Spread in a Connected Cell Population

Literature review for the fly_ovaries_processing_pipeline project

PubMed search conducted March 2026

This review addresses a specific analytical question: given that all VASA+ germ cells form a single connected component in the face-adjacency contact graph, how do we quantify whether they aggregate into a compact sphere-like mass, or whether they are dispersed throughout the tissue volume while maintaining a connection path between them? The review covers (1) graph-theoretic topology metrics applied to physical cell contact networks, (2) fractal dimension and lacunarity as shape descriptors for cell aggregates, and (3) geometric/spatial metrics derivable from cell centroid coordinates. For each metric, its interpretation, implementation in the existing pipeline, and required controls are described.

Biological question:
All VASA cells are in one connected component (confirmed by existing pipeline). But a single connected component can look very different: a dense ball of touching cells (compact) vs. a branching tree of cells strung together in chains (spread). The metrics below discriminate between these extremes and provide a scalar per sample for statistical comparison across conditions.

SECTION 1 - Graph Topology Metrics Applied to Physical Cell Contact Networks

Human Cancer Cell Radiation Response Investigated through Topological Analysis of 2D Cell Networks

Tirinato et al. - Annals of Biomedical Engineering 2023 | PMID: 37093401 | DOI: 10.1007/s10439-023-03215-z

Summary: Subjects 4 human cancer cell lines (H4, H460, PC3, T24) to photon radiation (0-6 Gy) and builds 2D cell contact graphs from fluorescence images. Computes clustering coefficient (cc), characteristic path length (cpl), and small-world coefficient (SW = cc/cpl ratio vs. random). Key result: higher radiation dose -> higher cc + lower cpl, meaning cells cluster tightly and are closer together in the graph. SW > 1 quantifies departure from random layout.

Relevance: MOST DIRECTLY RELEVANT. This is exactly the framework you need: applies graph metrics (cc, cpl, SW) to a physical cell contact network from microscopy images - the same network your pipeline builds. cc: how tightly are each cell's neighbors interconnected. cpl: average shortest path length across all cell pairs. Compact cluster = high cc + low cpl. Spread chain = low cc + high cpl. SW coefficient normalizes both relative to a random graph with the same N and degree sequence.

Display ideas: Bar charts of cc, cpl, SW per condition with error bars. Scatter plots cc vs. cpl coloring by condition. Correlation matrix of topology metrics vs. biological parameters.

Controls: Null: shuffle cell positions within each sample (preserve N and volume, randomize contacts). Compare SW to SW_random = 1 by definition. Dose-response curve serves as biological positive control.

Human lung-cancer-cell radioresistance investigated through 2D network topology

Tirinato et al. - Scientific Reports 2022 | PMID: 35902618 | DOI: 10.1038/s41598-022-17018-0

Summary: Predecessor paper from same group. Applies cc, cpl, and SW to 2D cell networks of H460, A549, Calu-1 lung cancer lines at 0-8 Gy radiation. Defines SW > 1 as evidence of 'discontinuous and inhomogeneous cell spatial layout' (spread). Shows that SW is sensitive to radiation dose and cell line radioresistance.

Relevance: Demonstrates that the SW coefficient is a sensitive, interpretable scalar for compactness/spread of a cell population from a 2D contact graph. Provides the methodological detail: graph built from Delaunay triangulation of cell centroids, edge added if cells are within a proximity threshold. Your face-adjacency graph is more biologically grounded (cells must physically touch) - a stricter criterion that should yield more meaningful cc and cpl values.

Display ideas: SW vs. dose curves per cell line. Heatmap of cc x cpl space per sample.

Controls: Random graph with same N and edge count as null. Varying proximity threshold as sensitivity analysis.

Collective dynamics of 'small-world' networks

Watts & Strogatz - Nature 1998 | PMID: 9623998 | DOI: 10.1038/30918

Summary: Foundational paper defining small-world networks. Regular lattices: high cc, high cpl. Random graphs: low cc, low cpl. Small-world networks: high cc AND low cpl simultaneously. The small-world coefficient SW = (cc_obs/cc_rand) / (cpl_obs/cpl_rand) > 1 identifies small-world topology. Demonstrates in C. elegans neural network, power grid, and actor network.

Relevance: Provides the foundational definitions for cc, cpl, and SW. Essential reference when reporting these metrics. Critically: a compact cell cluster resembles a regular lattice (high cc, moderate cpl); a spread chain-like cluster resembles a random graph but with lower cc and high cpl. This paper gives you the vocabulary to interpret your cell network topology.

Display ideas: The classic cc/cpl vs. rewiring probability plot (Figure 2 of Watts & Strogatz). Adapt for cell data.

Controls: Compare against equivalent random and regular lattice reference graphs.

SECTION 2 - Fractal Dimension and Lacunarity for Cell Aggregate Geometry

Aggregation dynamics of active cells on non-adhesive substrate

Mukhopadhyay & De - Physical Biology 2019 | PMID: 31042683 | DOI: 10.1088/1478-3975/ab1e76

Summary: Cellular automata model of self-assembling cell aggregates on non-adhesive substrate. Directly tracks: (1) compactness of aggregates as a scalar over time, defined as ratio of boundary cells to total cells; (2) fractal dimension of the growing aggregate boundary by box-counting - D=2 for compact filled disc, D<2 for rough/branching boundary; (3) cohesive strength (related to edge density in contact graph). Model captures transition from dispersed single cells to compact tissue structures.

Relevance: Directly defines 'compactness' as boundary-to-total-cell ratio for a connected cell aggregate - directly applicable to your 3D label images. In 3D: compactness = surface cells / total cells (cells with at least one non-cell neighbor). Compact ball -> small ratio. Spread structure -> large ratio. Fractal dimension of the outer surface boundary is also computable from your label image.

Display ideas: Time series of compactness and FD during aggregate growth. Phase diagrams of cohesion vs. compactness.

Controls: Simulated random aggregate as null. Compactness = 1 for a linear chain (all boundary), <1 for filled sphere.

Lung cancer - a fractal viewpoint

Lennon et al. - Nature Reviews Clinical Oncology 2015 | PMID: 26169924 | DOI: 10.1038/nrclinonc.2015.108

Summary: Review of fractal dimension (FD) and lacunarity for lung tumor morphology in CT/PET images. FD (box-counting): measures how a structure fills space across scales. FD=3 for solid sphere, FD<3 for branching/filamentous. Lacunarity: complementary measure of the 'gappiness' or texture of spatial distribution - how uniformly cells fill their bounding volume. High lacunarity = heterogeneous, gappy distribution. Low lacunarity = homogeneous filling.

Relevance: Introduces LACUNARITY as a key complement to FD. For a compact cell cluster: low lacunarity (cells fill space uniformly, few gaps). For a spread cluster: high lacunarity (large gaps between cell chains). Lacunarity is computed from a gliding-box algorithm on the binary label image - directly applicable to your 3D VASA label images. The combination FD + lacunarity gives a 2D descriptor space for cluster morphology that is more informative than either alone.

Display ideas: FD vs. lacunarity scatter plot per sample, colored by condition. FD + lacunarity together on bivariate plot.

Controls: Simulated solid sphere (high FD, low lacunarity) vs. random point cloud (low FD, high lacunarity) as reference.

Fractal dimension and lacunarity measures of glioma subcomponents are discriminative of grade and IDH status

Yadav et al. - NMR in Biomedicine 2024 | PMID: 39367752 | DOI: 10.1002/nbm.5272

Summary: Applies 3D fractal dimension (FD3D) and 3D lacunarity (Lac3D) to glioma subcomponents from MRI. High-grade gliomas: higher FD (more irregular, complex geometry) + lower lacunarity (denser tumor) in the enhancing region. IDH-mutant vs. wild-type show distinct FD + lacunarity signatures. Machine learning on FD3D alone discriminates grade with 97.9% sensitivity.

Relevance: Validates the 3D FD + lacunarity framework in volumetric imaging (analogous to your 3D confocal stacks). Key conceptual point: higher FD = more complex, space-filling (compact); lower FD = simpler, branching/filamentous (spread). Lower lacunarity = more uniform filling (compact). Both metrics are calculable from your 3D VASA binary label image using skimage.measure.

Display ideas: FD3D vs. Lac3D 2D scatter per sample. Box plots of FD3D and Lac3D by condition. ML classifier ROC.

Controls: Solid sphere and random point cloud controls. Cross-cohort validation (TCGA + UCSF datasets). Permutation test on FD values.

Fractals in pathology

Cross - Journal of Pathology 1997 | PMID: 9227334 | DOI: 10.1002/(SICI)1096-9896(199705)182:1<1::AID-PATH808>3.0.CO;2-B

Summary: Foundational review. Fractal dimension measures 'space-filling properties.' Box-counting method: cover structure with boxes of side r, count N(r), FD = -log(N(r))/log(r). Applications: tumor boundary irregularity, vascular branching, neuronal arborization. Key insight: FD quantifies structural complexity at multiple scales simultaneously.

Relevance: Essential methodological reference for box-counting FD. For a 3D cell cluster: FD=3 means cells fill a 3D volume (compact ball). FD=2 means cells form a sheet-like surface. FD=1 means cells form a linear chain. Real clusters will fall between these ideals. The box-counting algorithm is easy to implement on your 3D label image.

Display ideas: Log-log plot of N(r) vs. r (slope = FD). This is the standard quality-control plot for FD measurement.

Controls: Measure FD on a simulated solid sphere and on a random walk to bracket the expected range.

SECTION 3 - Geometric Metrics from Cell Centroid Coordinates

Characterization and classification of tumor lesions using fractal-based texture analysis and SVMs in mammograms

Guo et al. - International Journal of Computer Assisted Radiology and Surgery 2008 | PMID: 20033598 | DOI: 10.1007/s11548-008-0276-8

Summary: Compares 5 fractal dimension estimation methods for mammographic mass lesions. Key finding: FD of mass lesions is significantly LOWER than normal parenchyma (less space-filling). Lacunarity of mass lesions is HIGHER than normal (more gappy/heterogeneous). Combining FD + lacunarity yields highest classification performance (AUC = 0.90). Fractional Brownian motion (FBM) method outperforms standard box-counting for self-affine structures.

Relevance: Validates FD + lacunarity as paired descriptors of spatial distribution heterogeneity in biomedical images. Provides practical guidance: lacunarity as computed by gliding-box algorithm captures gap distribution at multiple scales. The FBM method may be worth comparing to box-counting for your 3D data.

Display ideas: ROC curves comparing FD alone vs. FD + lacunarity. Scatter of FD vs. lacunarity per sample group.

Controls: 5-fold cross-validation for classifier. Multiple FD methods compared (sensitivity analysis).

SECTION 4 - Metric Catalog: What to Compute and How

The following metrics are ordered from easiest to implement (using existing pipeline code) to most complex. All can be calculated from data already produced by the pipeline. Graph metrics require only the NetworkX contact graph. Spatial metrics require cell centroid coordinates. Fractal metrics require the 3D binary label image of the connected component.

A. Graph Topology Metrics (from existing NetworkX contact graph)

Characteristic Path Length (CPL) - Average shortest path

Formula / method: CPL = mean of all pairwise shortest paths: (1/n(n-1)) * sum_uv d(u,v). nx.average_shortest_path_length(G).

Interpretation: Compact ball: small CPL (cells across the cluster are few hops apart). Chain / branching: large CPL. Single number per sample -> directly comparable across conditions.

Pipeline: Requires connected component (already verified). Add to neighbor_summary CSV. O(n^3) runtime.

Effort: Easy - one NetworkX call. Slow for N > 2000; use nx.single_source_shortest_path_length with sampling.

Graph Diameter

Formula / method: Diameter = max of all pairwise shortest paths: max_uv d(u,v). nx.diameter(G).

Interpretation: The longest 'shortest journey' between any two cells. Compact = small diameter. A linear chain of N cells has diameter N-1. A complete graph has diameter 1.

Pipeline: Add to neighbor_summary CSV alongside CPL.

Effort: Easy - one NetworkX call. Same runtime concern as CPL.

Small-World Coefficient (SW)

Formula / method: SW = (CC_obs / CC_rand) / (CPL_obs / CPL_rand). CC_rand = 2*E / (N*(N-1)) (expected for random graph). CPL_rand = ln(N) / ln(mean_degree). SW > 1: more clustered + shorter paths than random -> compact/small-world. SW < 1: less clustered + longer paths than random -> spread/chain-like.

Interpretation: Normalizes CC and CPL relative to a random graph with the same N and edge count. SW > 1 is the fingerprint of a compact, locally well-connected cluster. SW < 1 means cells form long chains with few redundant connections (spread).

Pipeline: CC already computed. Add CPL. Compute random graph references analytically or via nx.erdos_renyi_graph(N, p) with p = 2E/(N*(N-1)). Compute SW per sample.

Effort: Easy once CPL is added. Reference values can be computed analytically.

Graph Eccentricity Distribution

Formula / method: Eccentricity of node u = max_v d(u,v). nx.eccentricity(G). Summary stats: mean eccentricity, radius = min(eccentricity), diameter = max(eccentricity).

Interpretation: Peripheral cells (far from the cluster center) have high eccentricity. Compact cluster: narrow eccentricity distribution (all cells are similarly central). Spread structure: wide distribution with a few very peripheral cells.

Pipeline: Use nx.eccentricity() - returns per-cell values. Map onto label image with paintLabelsByMetric.

Effort: Easy - already have the graph. Visualize as per-cell heatmap (eccentricity-painted labels).

Betweenness Centrality Distribution

Formula / method: BC(v) = fraction of all shortest paths that pass through node v. nx.betweenness_centrality(G, normalized=True).

Interpretation: In a chain-like cluster, a few bridge nodes have very high BC (bottlenecks). In a compact cluster, BC is low and uniformly distributed. Gini coefficient or coefficient of variation of BC distribution = single scalar for inequality.

Pipeline: Map BC onto label image to visualize bridge cells. Compute CV(BC) as compactness proxy.

Effort: Moderate - O(n^3). For large graphs use approximation: nx.betweenness_centrality(G, k=100).

B. Spatial Geometry Metrics (from cell centroid coordinates)

Radius of Gyration (Rg)

Formula / method: Rg = sqrt(mean_i |r_i - r_cm|^2), where r_cm = mean centroid of all cells. np.sqrt(np.mean(np.sum((centroids - centroid_mean)**2, axis=1))).

Interpretation: Mean RMS distance of cells from the cluster center of mass. Compact sphere: small Rg. Spread structure: large Rg. Normalize by expected Rg for a uniform sphere of same N and cell volume -> dimensionless index.

Pipeline: Cell centroids already in regionprops output. One line of numpy. Add to neighbor_summary CSV.

Effort: Trivial - pure numpy.

Asphericity / Anisotropy (from gyration tensor)

Formula / method: Build gyration tensor T_ab = (1/N) sum_i (r_ia - r_a_cm)(r_ib - r_b_cm). Eigenvalues l1 >= l2 >= l3. Asphericity = l1 - (l2+l3)/2. Anisotropy = (l1-l3)/(l1+l2+l3). np.linalg.eigh(gyration_tensor).

Interpretation: l1 = l2 = l3: perfect sphere (compact). l1 >> l2, l3: elongated cigar (spread in one direction). l1 = l2 >> l3: flat disc. Asphericity = 0 for sphere, > 0 for elongated/flat. These metrics detect directionality of spreading, not just total spread.

Pipeline: Add after centroid extraction. Yields 3 scalars: asphericity, anisotropy, shape parameter.

Effort: Easy - 10 lines of numpy.

Convex Hull Volume Ratio

Formula / method: Convex hull of all cell centroids: from scipy.spatial import ConvexHull; hull = ConvexHull(centroids). Ratio = sum(cell_volumes) / hull.volume.

Interpretation: Fraction of the convex hull volume actually occupied by cells. Compact sphere: ratio near 1 (cells fill the hull). Spread branching structure: ratio near 0 (large empty hull volume with sparse cells inside).

Pipeline: Cell centroids and volumes from regionprops. scipy.spatial.ConvexHull available in environment.

Effort: Easy. Note: ConvexHull degenerates for collinear/coplanar sets - add jitter guard.

Radial Cell Density Profile

Formula / method: Compute distances of all cell centroids from cluster centroid. Bin by radial distance (e.g. 2 um bins) -> N(r) per bin. Normalize by shell volume (4 pi r^2 dr) -> local density rho(r).

Interpretation: Compact spherical cluster: rho(r) peaks near r=0 then drops to zero at the cluster edge. Spread cluster: rho(r) is flatter, or peaks at a non-zero r (ring-like), or has a long tail. Full profile per sample -> scalar summary: e.g., half-radius R50 (radius containing 50% of cells).

Pipeline: Pure numpy/scipy. Produces a curve analogous to Ripley's K but for own-cluster density.

Effort: Easy. R50, R80 are compact scalars for statistical comparison across conditions.

C. Fractal Dimension and Lacunarity (from 3D binary label image)

3D Box-Counting Fractal Dimension (FD)

Formula / method: Binarize 3D VASA label image to 1=cell, 0=background. Cover with boxes of side r (e.g. r = 2, 4, 8, 16, 32 voxels). Count N(r) = number of boxes containing at least one cell voxel. FD = -slope of log(N(r)) vs. log(r). skimage.measure provides tools; custom box-counting ~20 lines.

Interpretation: FD = 3: cells fill a solid 3D volume (maximally compact). FD ~ 2: cells form a shell or surface (hollow ball or flat sheet). FD ~ 1: cells form a linear chain (maximally spread). Real clusters: FD between 2 and 3. Higher = more compact.

Pipeline: Applies to the 3D VASA refined label image (_refined.tif), binarized. Isotropic resampling first (already done at 0.25 um isotropic in the pipeline). Log-log linearity check (R^2 > 0.99) required for valid FD.

Effort: Moderate - implement box-counting loop (20-30 lines). Add FD and R^2 to summary CSV.

Lacunarity

Formula / method: Gliding-box algorithm: slide a box of side r through the 3D binary image. For each position, count S = sum of occupied voxels in box (0 to r^3). Compute Q(S,r) = probability distribution of S. Lacunarity L(r) = E[S^2] / E[S]^2 = (mean(S^2)) / (mean(S))^2. Average L(r) across scales for a scalar summary.

Interpretation: L(r) = 1: cells are uniformly distributed (translationally invariant). L(r) > 1: heterogeneous, gappy distribution. Compact cluster: low lacunarity (cells fill boxes uniformly). Spread cluster: high lacunarity (many empty boxes, some dense boxes). L(r) is scale-dependent -> plot L vs. r for full characterization.

Pipeline: Implement gliding-box on 3D binary label. Can subsample volume for speed. Scalar summary: mean L over a biologically relevant range of r (e.g. 2-20 um).

Effort: Moderate. The 3D gliding-box is O(n^3 * n_scales). Use stride > 1 for speed.

Surface-to-Volume Ratio of the Connected Component

Formula / method: Already computed per cell: surface area and volume from marching cubes (skimage). For the WHOLE connected component: sum of exposed surface (faces not shared with another VASA cell) divided by total cell volume. Equivalently: surface_area_component / volume_component.

Interpretation: Compact sphere: S/V = 4pi*r^2 / (4/3 pi r^3) = 3/r (minimal S/V for given volume). Spread structure: S/V is much larger (more exposed surface per unit volume). S/V ~ compactness in 3D - directly related to Mukhopadhyay's 'boundary cell ratio' metric.

Pipeline: Largely already computed per cell. Aggregate to whole-component: S_component = sum of per-cell faces NOT shared with another VASA cell (external faces only). V_component = sum of all VASA cell volumes.

Effort: Easy - extend existing regionprops extraction. One nested loop over face-adjacency graph.

SECTION 5 - Summary Comparison Table

Summary of all metrics, ordered by implementation effort. All can be added to the existing spatial analysis pipeline with minimal refactoring.

Metric	Compact value	Spread value	Data needed	Effort
Graph diameter	Small (few hops)	Large (many hops)	Contact graph (G)	Easy - nx.diameter(G)
Char. path length (CPL)	Low	High	Contact graph (G)	Easy - nx.avg_shortest_path_length
Small-world coeff. (SW)	SW > 1	SW < 1	G + degree sequence	Easy - formula after CPL
Eccentricity distribution	Narrow, low mean	Wide, high mean	Contact graph (G)	Easy - nx.eccentricity
Betweenness centrality CV	Low CV (uniform)	High CV (bottlenecks)	Contact graph (G)	Moderate - O(n^3)
Radius of gyration (Rg)	Small	Large	Centroids (regionprops)	Trivial - numpy
Asphericity / anisotropy	Near 0	Large	Centroids	Easy - numpy.linalg
Convex hull ratio	Near 1.0	Near 0	Centroids + volumes	Easy - scipy.spatial
Radial density R50	Small	Large	Centroids	Easy - numpy
Surface/volume ratio	Low (3/r for sphere)	High	Label image + adjacency	Moderate - extend existing
Box-counting FD	Near 3.0	Near 1-2	3D binary label	Moderate - implement loop
Lacunarity L(r)	Low (near 1)	High (>1)	3D binary label	Moderate - gliding box

SECTION 6 - Recommended Implementation Order and Controls

Recommended order: start with graph metrics (already have the graph) + geometric metrics (already have centroids). Add fractal metrics last. Use multiple metrics together - they capture different aspects of compactness.

Step 1 (immediate): Add CPL + diameter to existing pipeline

In analyzeVASANeighbors(), after graph construction: add nx.average_shortest_path_length(G) and nx.diameter(G) to the summary dict. Also add SW = (cc_mean/cc_rand) / (cpl/cpl_rand) where cc_rand = 2*E/(N*(N-1)) and cpl_rand = ln(N)/ln(k_mean). Zero additional dependencies.

Step 2 (easy): Add geometric metrics from centroids

From regionprops (already available): extract centroid array. Compute Rg, gyration tensor eigenvalues (asphericity, anisotropy), convex hull ratio (scipy.spatial.ConvexHull), radial density profile and R50. Add all to neighbor_summary CSV.

Step 3 (moderate): Add eccentricity + betweenness centrality

nx.eccentricity(G) and nx.betweenness_centrality(G). Use paintLabelsByMetric to visualize eccentricity and BC per cell - highlights peripheral and bridge cells visually. For large N, use approximated betweenness (k=100 sampling).

Step 4 (moderate): Add surface/volume ratio for whole component

Extend processVASASegmentationMasks: identify 'external' cell faces (faces bordering background or TJ cells, not shared VASA-VASA faces). S_external = sum of exposed face areas. S/V_component = S_external / sum(cell_volumes). Already have per-cell surfaces from marching cubes.

Step 5 (project): Box-counting FD + lacunarity

Implement on the 3D refined label image. Binarize, run box-counting at 6-8 scales, fit log-log line. Run gliding-box lacunarity at same scales. Add FD, L_mean, and log-log R^2 (quality flag) to summary CSV.

Critical controls for compactness metrics

*** Cell number (N) as covariate:** All graph metrics (CPL, diameter, SW) depend on N. Larger clusters mechanically have larger diameter. Always report N alongside topology metrics. If comparing conditions with different N, normalize or correct for N: e.g., CPL / ln(N) removes the baseline N-dependence.

* **Cell density as covariate:** In denser tissue, cells are mechanically forced to touch more neighbors, increasing CC and decreasing CPL. Report mean_degree alongside CC and CPL.
* **Null model - configuration model:** Your pipeline already implements configuration model permutations for CC. Extend to also permute for CPL and SW: build 1000 random graphs preserving degree sequence, compute CPL_null distribution, test whether observed CPL is outside the null envelope.
* **Null model - random spatial placement:** For geometric metrics (Rg, S/V, FD, lacunarity): place N cells randomly in the same bounding volume. This gives the expected values for a completely dispersed cluster and defines the 'maximally spread' reference.
* **Geometric shape of the tissue as reference:** The overall ovary geometry constrains how spread the cell population can be. Normalize Rg and convex hull ratio by the tissue bounding volume or convex hull.

Reference List

Tirinato L et al. (2023). Human Cancer Cell Radiation Response Investigated through Topological Analysis of 2D Cell Networks.
Annals of Biomedical Engineering 51(8):1859-1871. DOI: 10.1007/s10439-023-03215-z | PMID: 37093401

Tirinato L et al. (2022). Human lung-cancer-cell radioresistance investigated through 2D network topology.
Scientific Reports 12(1):12980. DOI: 10.1038/s41598-022-17018-0 | PMID: 35902618

Watts DJ & Strogatz SH (1998). Collective dynamics of 'small-world' networks.
Nature 393(6684):440-442. DOI: 10.1038/30918 | PMID: 9623998

Mukhopadhyay D & De R (2019). Aggregation dynamics of active cells on non-adhesive substrate.
Physical Biology 16(4):046006. DOI: 10.1088/1478-3975/ab1e76 | PMID: 31042683

Lennon FE et al. (2015). Lung cancer - a fractal viewpoint.
Nature Reviews Clinical Oncology 12(11):664-675. DOI: 10.1038/nrclinonc.2015.108 | PMID: 26169924

Yadav N et al. (2024). Fractal dimension and lacunarity measures of glioma subcomponents are discriminative of grade and IDH status.
NMR in Biomedicine 37(12):e5272. DOI: 10.1002/nbm.5272 | PMID: 39367752

Cross SS (1997). Fractals in pathology.
Journal of Pathology 182(1):1-8. DOI: 10.1002/(SICI)1096-9896(199705)182:1<1::AID-PATH808>3.0.CO;2-B | PMID: 9227334

Guo Q et al. (2008). Characterization and classification of tumor lesions using computerized fractal-based texture analysis and support vector machines in digital mammograms.
International Journal of Computer Assisted Radiology and Surgery 4(1):11-25. DOI: 10.1007/s11548-008-0276-8 | PMID: 20033598